

GENERAL SPECIFICATION G4.51

Computational Fluid Dynamics (CFD), Sump Modelling and Surge Analysis

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G4.51 COMPUTATIONAL FLUID DYNAMICS (CFD), SUMP MODELLING AND SURGE ANALYSIS

This specification covers the general requirement for the Computational Fluid Dynamics (CFD) study, sump modelling and surge analysis for the areas specified in the Particular Specification. These are collectively referred to as the Fluid Dynamics Analyses hereinafter unless requirements pertaining a particular study, modelling or analysis necessitates mentioning each individually.

G4.51.1 Submittals

The Contractor shall submit the following reports as minimum:

- (a) Preliminary Report
 - (i) After satisfactory completion of the Fluid Dynamics Analyses including all associated witness tests, the Contractor shall forward to the S.O.. Six (6) copies of a Preliminary Report showing the initial design and full details of any proposed modification. This shall consist of a written description of hydraulic phenomena leading to the proposed modifications and a description of the modifications themselves, together with fully dimensioned drawings. Upon acceptance, reinforced concrete or other structural design and calculations based on the recommendations of the report shall be submitted to the S.O..
- (b) Final Report
 - (i) The Contractor shall submit six (6) copies of the Final Report of the Fluid Dynamics Analyses to the S.O. after addressing and incorporating the S.O.'s comments. The Final Report shall consist of the following sections as minimum:
 - 1) Introduction;
 - 2) Terms of Reference;
 - 3) Description of the hydraulic system;
 - 4) Applicable assumptions;
 - 5) Statement of choice of scale and justification;
 - 6) Statement of all other criteria and bases for calculation;
 - 7) Description of the Fluid Dynamics Analyses and experimental procedure;
 - 8) Fully documented test results and calculations;
 - 9) Descriptions of tests carried out and modifications made;
 - 10) Fully dimensioned sketches, schematic diagrams and drawings showing the Fluid Dynamics Analyses including model construction and modifications made;
 - 11) Full photographic records of model construction and hydraulic conditions; and
 - 12) Discussion of results, conclusion, and recommendation including the following as minimum:
 - Hydraulic Appraisal/Engineering Discussion of the system design and operation;

- System characteristics and determination of steady flowrate for the full range of operating conditions;
 - Detailed explanation with the engineering significance of all computer results of transient simulations;
 - Selected pressure and flow graphs at salient locations along the main(s) plus a maximum and minimum pressure/distance envelope for each transient simulation if required; and
 - Detailed recommendations with outline sketches for any modification work to structure or any surge suppression equipment.
- (ii) The Contractor shall also submit six (6) copies of a suitably edited video recording showing the performance of the model at key flows and levels before and after any modifications, with an audio commentary to suit a non-specialist audience. The audio commentary shall be in English.

G4.51.2 Qualification

- (a) The Contractor or his specialist sub-contractor shall furnish track records of proven experience and capability of conducting the Fluid Dynamics Analyses in similar size projects. The track records shall have at least five (5) projects reference with the same applications to demonstrate commendable performance for delivering the sump models and surge analyses.

G4.51.3 General Requirements

- (a) The CFD study shall evaluate the proposed design of the pumping station, pipework, sump, wet-well and/or tanks under all operating conditions. The CFD study shall include an assessment of the risk of vortex formation that may be harmful to the pumps, recommendations for any modifications as needed for proper pump operation and a detail analysis of the flow conditions at pump inlets. The criteria for determining whether the approach flow to the pumps is suitable for operation will be the American National Standard for Pump Intake Design (ANSI/HI 9.8).
- (b) The Contractor shall demonstrate the specialist's experience and capability to carry out the modelling by citing experience in similar projects and shall identify at least five (5) supporting project references.
- (c) The scope of work shall include CAD Modelling, Grid Generation, Case Set-Ups, Simulations, Results Analysis and Reporting. The proposed CFD study shall include modelling of a short stretch of the inlet piping, the entire intake structure, the wet well and the suction piping with the pumping suction inlet nozzle to the plane of impeller. Each pump station being analysed shall have a minimum of three operating conditions tested to cover the peak, average and minimum flow scenarios.
- (d) The sump modellings and surge analyses shall include the full range of potential flow conditions including flow transients, critical flow characteristics, any vortex generation, material accumulation and any hydraulic deficiency.
- (e) The hydraulic modellings shall include sensitivity study to define influences of changes on boundary conditions, system parameters and pressure fluctuations in pipes under flow range of flow conditions.

- (f) The results shall indicate any modification desirable to give optimum operating conditions. Where modifications are required, further CFD modelling analyses, sump model tests and surge analyses corresponding to the modification shall be carried out to obtain the desired results.
- (g) Sump model tests shall also be carried out using suitable materials to simulate the patterns of flow, and any accumulation of materials in the sump, baffle walls, benching, and channels, etc. with and without modification.
- (h) Modelling shall confirm that the transient hydraulic behaviour, including pressure fluctuations and the range of water surface elevations in open chambers, wells, etc. conform to within acceptable limits.
- (i) The surge analysis for the surge pressure in the pumping mains application shall include the minimum, average and peak flows for all phases from initial and ultimate flow conditions and all possible combinations of pumps operation configurations as specified in the Particular Specification.
- (j) The Contractor shall present the results of the CFD modelling analyses, sump model tests and surge analyses and shall consult the S.O. regarding any necessary design modification to the wet well or sump structure, whichever is applicable and regarding any incorporation of surge suppression devices.
- (k) On acceptance of the reports and recommendations to achieve ideal hydraulic conditions, the Contractor shall submit detailed specifications and installation drawings on the surge pressure control equipment and modification on the civil and structural work not limited to infill, benching, baffling, vortex prevention cones or optimisation of pump location in the wet well or sump to the S.O. for approval. The cost for the modification work and providing the surge mitigation measure including the surge suppression system shall be deemed to have been included in the tender price with no further additional cost to the Board.
- (l) The surge suppression system, if required, shall consider the following options:
 - (i) Surge vessels with compressed air system;
 - (ii) Air Valves;
 - (iii) Flywheels on the pumps; and
 - (iv) Closing speed control of non-return valves.

G4.51.4 Sump Modelling

- (a) The objective of the sump modelling tests is to demonstrate that flow at the vicinity of the pumps inlets are of conducive flow conditions and patterns for optimum pump and hydraulic performance under the various combination of pump numbers in operation simultaneously as specified to meet the requirements of the various demand scenarios.
- (b) The sump modelling tests shall also assess the potential and impact of air entrainment in the fluid flowing into the pumps for all flow demand scenarios. Based on the test results, mitigation measures shall be recommended to eliminate air entrainment as far as possible. Where appropriate, these may include structural modification of the wet well such as inclined slab surfaces to receive flow any other suitable measure.

- (c) The required sump modelling tests shall be based on the CFD modelling.
- (d) Sump modelling shall involve the construction of physical sump models and scaled down setup to simulate operation of the sump and suction headers.
- (e) The model(s) construction shall comply with the laws of dynamic and geometric similitude.
- (f) Modelling Techniques: The numerical code employed in the investigation will be 'FLUENT' or equivalent. The simulations will be fully three-dimensional and will employ second order discretisation of Navier-Stokes equations using two-equation k-epsilon modelling of turbulence.
- (g) Geometry 3D model of the pump station will be generated in accordance with the specified drawing.
- (h) Pumps selection: As proposed by the Contractor complying with all applicable General and Particular Specifications.
- (i) Boundary conditions: All simulations will be done under steady state conditions, which mean that time variable will be eliminated from the basic flow equations. This is a simplified procedure, but most commonly used and shall be fully adequate to study flow conditions at intakes to pumping stations to represent predetermined critical conditions of constant water levels and constant flow rates. Under these circumstances the flow field will be represented by time-averaged values.

G4.51.5 Scaling Criteria for Wet Well or Sump Model

- (a) The wet well or sump model shall include the wet well or pump sump, whichever is applicable and extend upstream a sufficient distance to accurately demonstrate hydraulic conditions at the pump suction.
- (b) The model shall not be smaller than one-tenth full scale.
- (c) Surface finishes shall accord as closely as is practically possible with the theoretical requirements for representative boundary layer conditions.
- (d) The scale of the model shall be selected by considering, as a minimum, the following three principal non-dimensional parameters:
 - (i) The Froude number (F_r , ratio of inertial and gravitational forces);
 - (ii) The Reynolds number (R_e , ratio of inertial and viscous forces); and
 - (iii) The Weber number (W_b , ratio of inertial and surface tension forces).
- (e) The theory of scaling for the hydraulic model shall be based on the concept of dynamic similarity, i.e. the motion of the fluid particles in the model will be reduced in magnitude but will be similar in motion to the equivalent particles in the full size structure. The model shall be a direct scaling of the proposed structure and operated so that the fluid motion is dynamically similar, in order that parameters such as water level, velocity and flow direction may be directly scaled up to predict actual values in the full size structure.

- (f) The velocity ratio between model and full size structures shall maintain similar Froude numbers for both. Where tests are being carried out to determine vortex formation, a 1.5 times Froude flow may be used thus making the model Reynolds number closer to that of the full size structure where turbulence and boundary layer development are more closely reproduced.
- (g) The scale of the model shall be submitted and approved by the S.O. prior to construction of the model.
- (h) A photographic and dimensioned sketch record shall be made showing model construction and modifications.
- (i) A photographic record shall be made showing the hydraulic conditions at strategic flow rates and levels.

G4.51.6 Wet Well or Sump Model Construction

- (a) The main frame of the model shall be constructed from exterior grade plywood and/or galvanized sheet steel. For inspection purposes clear Perspex sheet shall be used to give a clear view of all areas of the model. Benching, infill and baffles shall be constructed in hardwood timber and for smaller areas putty or similar may be used.
- (b) All details (such as penstocks, stop-logs, pipes, and benching) shall be included in the model.
- (c) Modifications and additional benching, infill and baffles walls found necessary to achieve the desired hydraulic conditions shall, where practically possible, be of removable construction to allow the original model structure to be retained.
- (d) Where screens are required, the screens and their supports shall be modelled. It shall be sufficient to use a model screen of about the same blockage ratio as in the prototype.
- (e) The flow shall be provided by service pumps remote from the model. The test facility, pipework, valves, and flow meters shall be such as to enable all operational combinations of flow to be reproduced from the minimum (or trouble-free) water level to maximum level in the wet well or pump sump. Water inflow into the model shall be similarly controlled.
- (f) Each pump suction shall be modelled as closely as practically possible to the expected pump configuration and shall be fitted with free-rotating vanes at the point equivalent to the location of the pump impellers, in order to register swirl at these points.
- (g) The suction piping configuration or bellmouth, whichever is applicable shall be connected to pumps remote from the model via a system of pipework, valves and flowmeters that allow all operational combinations to be reproduced from the minimum (or trouble-free) water level to the maximum level in the sump. Water inflow into the model shall be similarly controlled.
- (h) The model shall be instrumented to enable flow control by means of manual sluice gates and from pumps. An orifice plate flowmeter may be used. Dye or other tracers shall be used to establish flow patterns.

- (i) Before construction of the model, scope for modifications at site, fixed boundaries, etc., shall be discussed with the S.O., and the model constructed accordingly. The facility for flow visualisation and photography shall be included at critical sections.

G4.51.7 Testing Criteria

- (a) Observation of any vortex formation should be classified by assigning a number (1 through 6) which corresponds to the vortex-type classification by Hydraulic Institution Standards (HI), or Alden Research Laboratory (ARL), Massachusetts, USA.
- (b) A vortex type 1 denotes the weakest class (surface swirl) and a type 6 vortex denotes the strongest class (full air-core to intake). A vortex drawing air to an intake can cause damaging vibrations and impair pump performance. If vortex action is observed during the testing, measures to suppress such activity shall be taken.
- (c) The allowable angle of swirl in the pump suction will depend upon the characteristics of the particular prototype pump being employed and is normally defined by the pump manufacturer. However, allowable angles of swirl typically lie in the range of 2° to 5°. For the purpose of these investigations, any swirl in excess of 5° will not be acceptable and measures to improve the situation shall be taken, unless otherwise agreed with the S.O..
- (d) Swirl angles in the model pump suction shall be measured using the rotor in the suction pipe, which shall preferably have straight vanes forming a cruciform.
- (e) The necessary or minimum submergence of the inlet pipe or bellmouth, whichever is applicable shall be on the basis of satisfying both the NPSH requirement of the pump and obtaining the required flows with no vortex formation.

G4.51.8 Witness Tests and Inspections

- (a) The Contractor shall undertake initial tests and document any initial problem observed. This documentation shall be made available to the S.O., prior to an initial witness test of the model.
- (b) An initial witness test shall then be undertaken during which possible problem areas and corrective modifications shall be discussed. The Contractor shall give at least two (2) weeks' advance notice of the availability of the model for witnessing the initial test.
- (c) The Contractor shall then conduct fully comprehensive tests on the model and make modifications where necessary. These tests shall be fully documented with drawings of any modifications made. This documentation shall be made available to the S.O., prior to a second witness test.
- (d) A second witness test shall then be undertaken in which the Contractor shall demonstrate the effectiveness of corrective modifications of which there shall preferably be more than one option.
- (e) After satisfactory completion of the second witness test, the Contractor shall forward the Preliminary Report to the S.O. for approval. The Preliminary Report shall be submitted no later than twelve (12) weeks after the Contract award, unless otherwise specified.

- (f) Within four (4) weeks of approval of the Preliminary Report by the S.O., the Contractor shall submit the Final Report of the model test to the S.O. after incorporating all comments. The Contractor and the sub-contracted specialist shall attend the meetings as required by the S.O. and present the reports for the preliminary and final stage of study to the Board and S.O. unless otherwise specified.
- (g) Upon implementing the modification work on the structure based on the approved Sump Model test final report and/or the surge suppression devices based on the approved Surge Analysis final report, the Contractor shall take measurements to establish the surge pressure in the system. All labour, expertise, and suitable equipment consisting of recorders, pressure gauges and all other necessary instruments shall be provided for the purpose of measuring the surge pressures in the pumping mains for the various possible operating conditions. These measurements shall be submitted to the S.O. and shall be compared with the results in the surge report.

G4.51.9 Surge Analysis

- (a) Surge Analysis shall be carried to determine the maximum positive and negative pressure experienced by the pumping mains, with recommendation if any surge protection equipment (eg. surge vessels with compressed air system, head tank, combination air valves, etc.) are required and whether the pipe material selected can withstand the above pressures.

G4.51.10 Fly Wheels

- (a) Flywheels, if recommended and approved by the S.O., shall be of cast iron to BS EN 1561 or other approved materials and shall be statically and dynamically balanced.
- (b) It shall be noted that the use of flywheel may affect the drive motors, electrical starters, motor stools and transmission shafts, etc. The changes and/or modifications required shall be carefully considered and all costs involved shall be deemed to have been included in the tender prices.

G4.51.11 Surge Vessel

The surge vessel shall meet the following additional requirements:

- (a) The surge vessel shall be designed to fully comply with the requirements of the Chief Inspector of Factories, Ministry of Manpower.
- (b) It shall be noted that in recommending the size and type of the surge vessel, its application (used water or clean water), the location and its surrounding shall be taken into consideration.
- (c) The compressed air type surge vessel shall be used for used water application unless otherwise specified.
- (d) The surge vessels and the associated structural design including the ladder, platform, walkways, supports shall be designed and endorsed by the relevant discipline P.E.. The design details shall be submitted to S.O. for approval.
- (e) Unless otherwise specified, each surge vessel shall be of the horizontal steel cylindrical type, with hemispherical ends.
- (f) No field welding will be allowed in the fabrication or installation of the surge vessels.

- (g) The installed surge vessel shall be tested to the requirements of the Chief Inspector of Factories, Ministry of Manpower and in the presence of the S.O.. All necessary equipment, etc. required for the tests shall be provided and all costs involved shall be deemed to have been included in the tender prices.
- (h) Laboratories accredited by the Singapore Accreditation Council (SAC) under the SINGLAS scheme shall carry out all mechanical and non-destructive testing for all pressure vessels, as required by the Chief Inspector of Factories, Ministry of Manpower. Only test reports issued by SAC-SINGLAS accredited laboratories shall be recognised.
- (i) The internal surfaces of the surge vessels shall be wire brushed, sand blasted and painted with three (3) coats of epoxy paint. This shall be done prior to delivery to site and may be inspected by the S.O. at the fabricator's workshop. External painting shall comply with General Specification on Painting and Protection.
- (j) The design, construction and testing of the vessel shall conform to the ASME Boiler and Pressure Vessel Code. Section VIII Unfired Pressure Vessels. Corrosion allowance for the shell and the dish ends shall be provided.
- (k) Each tank shall be of butt welded construction, amply proportioned and designed to withstand separately, or in combination, the loads and forces exerted by :-
 - (i) The weight of the tank;
 - (ii) An internal pressure equal to the Design Pressure; and
 - (iii) The weight of a full tank of water.
- (l) Welding shall be in accordance with the current ASME Code for Unfired Pressure Vessels or equivalent British Standard.
- (m) Butt welds shall be fully radiographed in accordance with the above code. Joints shall have a maximum joint efficiency of 100% of that of the connecting plates. All radiographs, including location plans and interpretations, shall be submitted to the S.O. for approval.
- (n) Branch-to-shell welds shall be magnafluxed before pressure testing.
- (o) Certificates shall be provided to confirm the results of shop fabrication procedures and welding tests as required by the ASME specification or BS EN 286-1:1998+A2:2005.
- (p) The following accessories shall be provided for each tank:
 - (i) The steel plate for the pressure vessel shall conform to ASTM A516 Grade 70 to "Fine Grain Practice", or an equivalent steel having good notch toughness unless otherwise approved by the S.O..
 - (ii) Each tank shall be adequately supported on steel saddles which are fixed permanently to the tanks. One tank support shall be suitable for rigid anchorage to the concrete floor. The others shall be of the sliding type, designed to allow for thermal expansion. Soleplates shall be provided with corrosion resistant shoes and stainless steel holding down bolts for the saddles to slide on. The structure of the tanks, support and anchor bolts shall be designed and endorsed by the relevant discipline P.E..

- (iii) Material for vortex plate, baffle plate, manhole, etc., where applicable shall be the same as specified for the pressure vessel.
- (iv) Access manholes of at least 600 mm in diameter shall be provided in the shell of the vessel at top and bottom.
- (v) 2 nos. of safety valves to BS 1123 or ASME approved spring-loaded pressure relief valves with stainless steel trim, suitable for air and water service, shall be provided. Each valve shall be set to relieve at pressures in excess of the design pressure. The valve shall not be less than 50 mm diameter. A test lever shall be provided with a chain pull for operation at ground level.
- (vi) A lubricated plug valve of the 3-way, 2 port, 2 position type rated to the same design pressure as the surge vessel but shall not be less than 800 kPa shall be furnished in order to select either of the two relief valves to service. Piping between tank and relief valves shall be provided.
- (vii) A sight-glass gauge shall be provided with a graduated scale and isolating lock which shall shut automatically in the event of breakage of the gauge glass. The gauge glass shall be protected with a sheet metal surround with the sides and back painted matt white. The front shall have 12 mm thick transparent armoured glass providing not less than 50 mm wide clear view of the glass gauge. The assembly shall facilitate easy dismantling for cleaning and removal of the gauge glass.
- (viii) Valved drainage pipework and compressed air pipework connections shall be provided.
- (ix) Pressure gauges of the Bourdon Type, not less than 100 mm dial diameter shall be provided, with black aluminum cases, stainless steel trim and suitably graduated to suit the test pressure rating of the surge vessels.
- (x) An orifice plate of stainless steel Type 316 construction, suitable for insertion between pipe flanges shall be provided.
- (xi) A structural aluminum ladder, maintenance platform with hatches and railings, and all other appurtenances required shall also be provided for each surge vessel.
- (xii) The Contractor shall register the tank, and obtain and provide with each tank a certificate of approval from the Ministry of Manpower, Singapore. These certificates shall be suitably framed for display in the pumping stations and the relevant details shall be painted on the vessels. This is applicable for all such items under this Contract.
- (xiii) A stainless steel SS316 plate of adequate size with embossed or engraved markings showing manufacturer, serial number, design pressure, design temperature, test pressure, volume, thickness of shell and heads, material of construction and date of fabrication shall be fixed to each surge vessel at a readily visible position.

G4.51.12 Results and Reports Submission

- (a) The Contractor shall submit a formal report for each of the CFD study, sump modelling test and surge analysis including the detailed write-out of the proposed modifications, if required, on the structural and mechanical configurations of the wet wells, well water levels, layout of pump units, etc. to achieve the flow patterns conducive for effective and efficient pump operation and performance. The report for the surge analysis shall also include the detailed write-out of the proposed surge mitigation measure(s). Each report shall comprise the following contents as a minimum:
 - (i) Abstract (summarising the purpose and objectives of the report);
 - (ii) Conditions and situations of the pumps/wet wells/suction piping (demonstrating the Contractor's appreciation and understanding of the system);
 - (iii) Theoretical background of CFD modelling test applicable to the concerned requirement;
 - (iv) Applicable assumptions;
 - (v) Scenarios being modelled and analysed (covering various number of pump duty units in operation and operating water levels, etc.);
 - (vi) Results of analysis consisting of tabulation summarising the analysis results including but not limited to the following:
 - 1) Observed phenomena (eg. surface or/and wall vortices);
 - 2) Associated problems;
 - 3) Graphs;
 - 4) Velocity plots;
 - 5) Flow pattern plots; and
 - 6) For the various scenarios including those following implementation of rectification measures.
 - (vii) Recommendations where applicable (including solutions or methods to overcome problems of vortex or foreseeable cavitation such as introduction of baffle walls, vortex prevention cones, back wall flow splitters, etc.);
 - (viii) Conclusions, including a confirmation statement that the recommendations and final proposed design will fulfil the required flow conditions and retention performance, or that the flow path to the pumps and pump operation will not result in the occurrence of vortex or turbulent flow which may cause serious pump cavitation or vibrations;
 - (ix) Where required by the recommendations of the result of analysis and modelling, the modifications to the wet wells' structures and pumps suction headers shall be included in the detailed design and construction;
 - (x) All costs arising from any required structural and mechanical design modification, construction and installation shall also be deemed to have been allowed for in the Contract Sum;

- (xi) Deviation from satisfactory performance; and
- (xii) Should the pumps installed according to the recommendations of the CFD study not perform satisfactory, the Contractor shall engage the CFD Specialist and experienced personnel to perform a detailed investigation for trouble-shooting. All costs associated with trouble-shooting and rectification measures shall have been included in the tender costs.

END OF SECTION